Component Selection & Analysis Report

Cancellation of Self-Sensing Due to Ego-Motion of Vibration-Based Mechanosensors

1. Computing Module — NVIDIA Jetson Nano

A capable onboard computing module is required to run the bio-inspired cancellation algorithm, acquire sensor data in real time, and handle any machine learning model inference. The module must offer sufficient CPU and GPU performance within the power and size constraints of the mobile platform. Three candidates were evaluated: the NVIDIA Jetson Nano, the Raspberry Pi 4B, and the NVIDIA Jetson Xavier NX.

Criterion	Jetson Nano (Chosen)	Raspberry Pi 4B
CPU	Quad-core ARM A57 @ 1.43 GHz	Quad-core ARM A72 @ 1.8 GHz
GPU	128-core Maxwell GPU	VideoCore VI (no CUDA)
RAM	4 GB LPDDR4	4 / 8 GB LPDDR4
CUDA Support	Yes	No
Power Draw	5–10 W	3–7 W
Cost	LKR 28,500.00	LKR 61,350.00
ROS Support	ROS / ROS2	ROS / ROS2

Figure 1: NVIDIA Jetson Nano

Figure 2: Raspberry Pi 4B

The Jetson Nano was selected because it provides GPU-accelerated computing with CUDA support, essential for running neural network-based cancellation models. The Raspberry Pi 4B is more power-efficient but lacks a CUDA-capable GPU.

3. Accelerometer / IMU — MPU6050

The accelerometer is the core sensing element of the robotic antenna and also serves as the reference ego-motion sensor mounted on the robot body. It must provide high sensitivity across the vibration frequency range of interest, a digital interface compatible with the Arduino Uno, and low-noise characteristics to enable detection of subtle contact-induced vibrations. Two devices were evaluated: the MPU6050 (6-DOF IMU), and the ADXL345.

Criterion	MPU6050 (Chosen)	ADXL345
DoF	6 (3-axis accel + 3-axis gyro)	3-axis accelerometer only
Interface	I2C / SPI	I2C / SPI
Acceleration Range	±2/±4/±8/±16 g	±2/±4/±8/±16 g
Gyroscope	Yes — 3-axis	No
Noise Density	400 µg/√Hz	150 µg/√Hz
Arduino Libraries	Well-documented	Good
Cost (LKR approx)	LKR 690.00(module)	LKR 630.00(module)

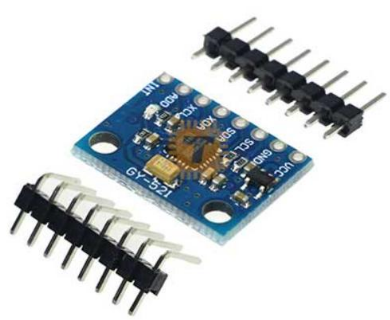

Figure 3: MPU6050 Accelerometer

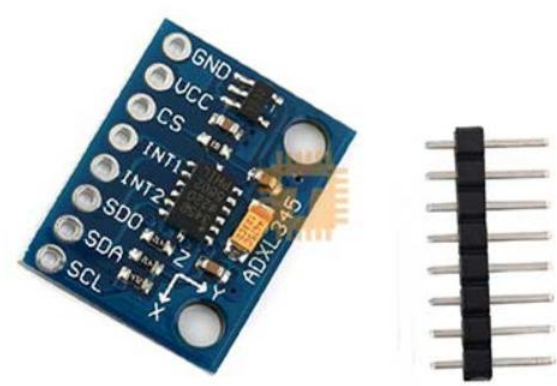

Figure 4: ADXL345 Accelerometer

The MPU6050 was selected for its combination of 6-DOF sensing (accelerometer + gyroscope), very low cost, and extensive Arduino library support. The gyroscope data provides additional rotational ego-motion information useful for the cancellation algorithm. The ADXL345 lacks a gyroscope, which would limit ego-motion characterisation to translational motion only.

4. Servo Motors — MG995

Servo motors drive the Pan-Tilt unit to orient the robotic antenna in different directions. They must provide adequate torque to hold the antenna position against vibration loads, angular precision for repeatable positioning, and compatibility with PWM control from the Arduino Uno. Three servos were considered: the MG995, the SG90, and the DS3218.

Criterion	MG995 (Chosen)	SG90	DS3218
Gear Type	Metal gear	Plastic gear	Metal gear
Stall Torque	10kg/cm @ 6V	1.6 kg/cm @ 4.8V	21.5 kg/cm @ 6.8V
Operating Voltage	4.8–7.2 V	4.8–6 V	4.8–6.8 V
Rotation Range	120° (standard)	180°	270°
Control	Standard PWM	Standard PWM	Standard PWM
Weight	55 g	9 g	60 g
Cost (LKR approx)	LKR 1150.00 (motor)	LKR 320.00 (motor)	LKR 4100.00 (motor)

Figure 5: MG995

Figure 6: DS3218

The MG995 servo was chosen because it has metal gears and enough torque (10 kg·cm) to keep the antenna fixed in place while the robot moves. This helps to avoid unwanted movement of the antenna and reduces extra noise in the signals. The SG90 servo is light and low cost, but it has plastic gears and very low torque, so it cannot properly support the antenna when vibrations occur. The DS3218 servo provides higher torque and a larger rotation range, but it is heavier and more expensive than needed for this project.

5. Antenna Tube — Poly-Acrylic Tube

The antenna tube is the mechanical element that transmits contact vibrations from the tip to the accelerometer. Its material properties, stiffness, density, and damping directly affect the vibration transmission characteristics and, therefore, the sensor signal quality. Three tube materials were evaluated: poly-acrylic (acrylic / PMMA), carbon fibre, and aluminium.

Criterion	Poly-Acrylic / PMMA (Chosen)	Carbon Fibre Tube	Aluminium Tube
Density	~1.18 g/cm³ — very light	~1.55 g/cm³ — light	~2.7 g/cm³ — moderate
Stiffness (E)	~3.2 GPa	~70+ GPa	~69 GPa
Internal Damping	Moderate	Low	Low
Machinability	Excellent — easily cut & drilled	Difficult — specialist tools	Good
Transparency	Yes — allows internal inspection	No	No
Cost (LKR approx)	LKR 330.00 per metre	-	-
Availability	Very high	Moderate	High

Figure 7: PMMA

Figure 8: Carbon Fibre

Figure 9: Aluminium Tube

Poly-acrylic (acrylic / PMMA) tubing was selected for the antenna structure due to its low mass, ease of machining, and good availability. Its moderate internal damping is acceptable for this application as the accelerometer at the tip will capture both contact and ego-motion signals, with the cancellation algorithm responsible for separation rather than passive mechanical filtering. Carbon fibre tubing would offer superior stiffness-to-weight but is difficult to cut and drill in a standard lab setting. Aluminium tubing is heavier and harder to mount the accelerometer on without specialist fixtures.

6. Battery — 12V Ni-MH Battery Pack

The power source should provide power to the robot drive motors (need 12V), servo motors, and the onboard computer module. It should be able to run the system continuously for experiments. Two battery types were considered for this purpose: Nickel-Metal Hydride (Ni-MH) batteries and Lithium Polymer (LiPo) batteries.

Criterion	**12V Ni-MH Pack (Chosen)**	**11.1V LiPo Pack**
Nominal Voltage	12V	11.1V (3S)
Energy Density	Moderate (~60–80 Wh/kg)	High (~150–250 Wh/kg)
Safety Risk	Low	Higher — flammable if damaged
Charge Management	Simple — standard charger	Requires balancing charger
Cycle Life	500–1000 cycles	300–500 cycles
Cost (LKR approx)	LKR 1720.00	LKR 6290.00
Lab Regulations	Generally permitted	May require special storage

The 12V Ni-MH battery pack was selected because it directly matches the voltage needed for the 4WD robot motors. It is also safer and easier to charge. In a university laboratory environment, Ni-MH batteries are usually preferred over LiPo batteries because they have a lower risk of fire and are easier to store. LiPo batteries can store more energy, but they need careful handling and special balancing chargers

7. Microcontroller & Motor Controller — Arduino Uno

The microcontroller is used to control low-level tasks such as generating PWM signals for the servo motors, controlling the motors of the 4WD robot platform, communicating with the MPU6050 sensor through I2C, and sending data to the Jetson Nano through serial communication. A motor controller module or shield is also needed to drive the 4WD motors using 12V power. Three microcontroller options were considered: Arduino Uno with an L298N motor controller, STM32F4 Discovery, and Arduino Mega.

Criterion	Arduino Uno + L298N (Chosen)	STM32F4 Discovery	Arduino Mega
Clock Speed	16 MHz	168 MHz	16 MHz
PWM Channels	6	16+	15
I2C Support	Yes	Yes	Yes
Motor Controller	L298N shield — 2A per channel	External driver required	L298N or similar
Programming	Arduino IDE	STM32CubeIDE	Arduino IDE
Community Support	Excellent	Moderate	Excellent
Cost (USD approx)	LKR 2250.00 (Arduino Uno) LKR. 420.00 (L298N)	LKR 4250.00	LKR 4850.00

Figure 10 : Arduino UNO

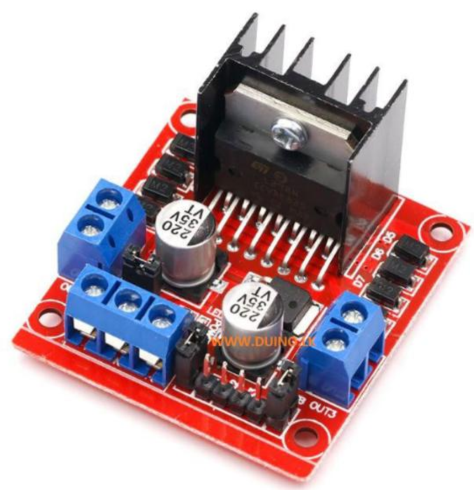

Figure 11 : L298N

The Arduino Uno with the L298N motor controller was selected because it is simple to use and has many community libraries available. It is also directly compatible with all selected components such as the MPU6050 sensor through I2C communication, servo motors through PWM signals, and the motor driver through digital and PWM pins. The Arduino IDE makes programming easier and reduces development time compared to STM32, which is more powerful but needs more learning and a more complex setup. The Arduino Mega has more input and output pins, but it is larger and more expensive, while not giving extra benefits for this project.

8. Mobile Robot Platform - 4WD Omni-Directional-Arduino-Compatible-Mobile-Robot

The mobile platform serves as the base of the entire system, carrying the Pan-Tilt unit, antenna, computing hardware, and battery. The platform must provide stable four-wheel drive for controlled ego-motion experiments, sufficient payload capacity for all components, and a rigid chassis to transmit representative vibration profiles to the antenna during motion trials.

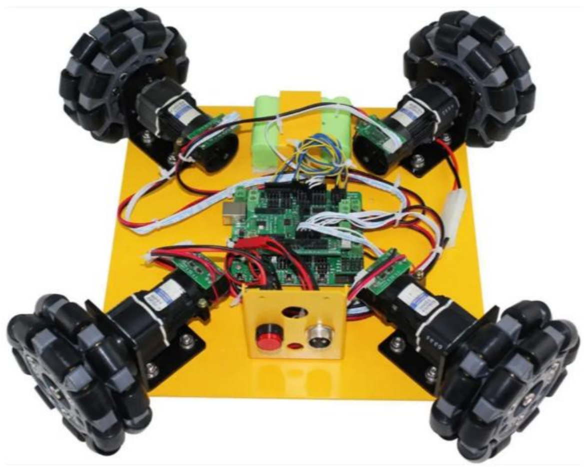

Figure 12: Mobile Robot Platform